From Scientific Test to Formula Discovery：A Survey of LLM-driven Symbolic Regression

1. **Team members and responsibilities:**
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2. **Background and motivation：**
   Data-driven scientific discovery has been called the fourth paradigm of science. Symbolic regression aims to automatically discover concise, interpretable mathematical formulas, for example， physical laws and biological models from observational data, and is a core task in AI for science. Classical symbolic regression, represented by genetic programming and sparse identification, is highly interpretable and capable of discovering novel formulas, but suffers from low search efficiency and difficulties in exploiting domain prior knowledge， presenting a rather disappointing role into the unknown.

   Since 2023 large language models have brought a new probability to this task and the direction now goes into the center of industry attention: AI for science has become a major focus for scientific discovery and OpenAI's math release, containing hundreds of AI generated mathematical results, shows how rapidly LLM mathematical and scientific reasoning is improving. LLM-SR reformulates formula discovery as code generation, letting LLMs program candidate formulas by scientific priors. And SymbolBench places symbolic regression and causal discovery side by side as core evaluation tasks for LM symbolic reasoning. The literature at this intersection is growing fast but remains fragmented, lacking a unified taxonomy and systematic comparison, which is exactly the gap this survey aims to fill.

3. **Problem definition and research questions:**
   Core question: What technical approaches exist for symbolic regression with LLMs and what are the capabilities and limitations of each approaches?

i. Into which categories can LLM-based symbolic regression methods be divided? What are the core ideas and representative methods of each?
ii. How do the approaches trade off in fitting accuracy, formula complexity, interpretability, data efficiency, and use of prior knowledge evaluation?
iii. Can existing benchmarks (SR bench, symbol bench) and metrics fairly evaluate LLM based methods? What are the shortcomings in assessing the reliability and general izability of machine-discovered formulas?
iv. What opportunities and challenges lies in prior injection? What if the scope goes wide, such as multimodal scientific data, reproducibility verification of machine discovery formulas and intersection with causal discovery?

4. **Research Scope & Preliminary Taxonomy**

Focus on the intersection of LLMs and symbolic regression from 2021 to 2026; classical

symbolic regression is covered only as background; the intersection with causal discovery is included in the discussion of evaluation and open problems.

4.1 Classification rationale. The taxonomy is organized by one criterion: **how the LLM participates in symbolic regression**. This yields four approach families that are mutually exclusive in their primary mechanism:

- **Approach 1 · LLM direct generation:** the LLM *is* the generator — formula discovery is reformulated as code generation/program synthesis, driven by prompt engineering and in-context learning (e.g., LLM-SR).
- **Approach 2 · LLM-guided evolutionary search:** the LLM *assists* an evolutionary algorithm — it proposes mutations and injects priors while genetic programming still performs the search (e.g., FunSearch).
- **Approach 3 · knowledge-prior injection:** the LLM sits *before or alongside* the search, mining formulas, terminology, and causal claims from scientific literature to constrain the search space (e.g., AI-Descartes).
- **Approach 4 · evaluation and benchmarks:** not a method family but the meta-layer — how all of the above are measured (SRBench, SymbolBench, and metrics for accuracy, complexity, and interpretability).

A background layer (classical SR: PySR, SINDy, end-to-end Transformers) precedes these four to establish what LLMs improve upon. Within each approach, methods are further organized along second-level dimensions: *degree of LLM involvement* (sole generator vs. assistant vs. prior provider), *how prior knowledge is represented* (in-context examples, prompt constraints, extracted facts, verifier feedback), and *task setting* (single-equation discovery, ODE/PDE systems, causal structure learning). This two-level structure makes the taxonomy both exhaustive for the primary mechanism and fine-grained enough for meaningful comparison.

4.2 Paper-to-category mapping (working example).

Approach 1 — direct generation: LLM-SR [8], and prior code-generation-style SR studies underlying it; SymbolBench's LLM-as-predictor setting [10].

Approach 2 — guided evolutionary search: FunSearch [7]; SymbolBench's LLM + genetic programming framework [10].

Approach 3 — prior injection: AI-Descartes [9]; literature-mining pipelines for formulas and causal claims [12, 14].

Approach 4 — evaluation: SRBench [3], SymbolBench [10], and the metric discussions in [4, 5, 6].

Background layer: SINDy [1], AI Feynman [2], end-to-end Transformers [4, 5], PySR [6].

（Assisted by AI with this Item）

5. References (preliminary list)

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